

Kai-Jie Lin

linkai0508@gmail.com | +886-965-457-866

Home Page | GitHub: KJLdefeated

Education

National Yang Ming Chiao Tung University (NYCU)

B.S. in Computer Science

Overall GPA: 3.83/4.3

Hsinchu, Taiwan

Sep. 2021 – Jun. 2025

University of Illinois Urbana-Champaign (UIUC)

Exchange Student in Computer Science

Illinois, USA

Aug. 2024 – Dec. 2024

Experience

NYCU Reinforcement Learning and Bandits Lab

Undergraduate Researcher

Hsinchu, Taiwan

Jan. 2023 – Present

- Currently developing a gradient-based Q-Transformer module that augments large language models with explicit reasoning in a latent concept space, enabling step-by-step credit assignment and non-autoregressive sequence generation beyond traditional token-by-token decoding.
- Proposed a Transformer-based Q-learning algorithm for Multi-Objective Bayesian Optimization that addresses myopic acquisition strategies by learning non-myopic, history-dependent policies.
- Implemented the framework and built large-scale experimental pipelines, achieving 20% hypervolume improvements over SOTA methods on synthetic benchmarks and real-world 3D Gaussian Splatting tasks.

Rayleigh Vision Intelligence (RVI)

AI Software Engineer (Part-Time)

Hsinchu, Taiwan

Feb. 2025 – Present

- Boosted Micro-LED wafer transfer efficiency by 75% by deploying a reinforcement learning based scheduling system.
- Built a horizontally sharded PostgreSQL database managing 1B+ records with <100ms query latency, enabling real-time manufacturing analytics.
- Developed a Model Context Protocol (MCP) framework connecting LLMs to internal databases and tools, enabling scalable AI-assisted workflows.

Reinforcement Learning (Graduate Course)

Teaching Assistant

Hsinchu, Taiwan

Spring 2024/2025

- Supported course delivery for 80+ students, spanning theoretical foundations and practical applications of reinforcement learning.
- Advised seven student teams (each with 3-4 members) on research directions for their implementation and theory projects.
- Designed and graded theoretical problem sets and programming assignments (Python, PyTorch), ensuring fair and timely evaluation.

Publications

BOFormer: A Generalized Deep Q-Learning Framework for Multi-Objective Bayesian Optimization

Yu-Heng Hung*, **Kai-Jie Lin***, Yu-Heng Lin, Chien-Yi Wang, Cheng Sun, Ping-Chun Hsieh.

International Conference on Learning Representations (ICLR), 2025. (* equal contribution)

Selected Projects

GPU Kernel Optimizations

UIUC, 2024

Course Project, Applied Parallel Programming (Graduated-level)

- Achieved 75% of cuBLAS performance in matrix multiplication with shared-memory tiling and Tensor Core acceleration.
- Accelerated convolution kernels using kernel fusion and CUDA streams to overlap computation with data transfer, delivering $2.5\times$ faster execution.
- Competed in the course-wide CUDA kernel optimization challenge, ranking in the top 10% of students for performance on convolution benchmarks.

Statistical Reinforcement Learning Project

UIUC, 2024

Course Project, Statistical Reinforcement Learning (PhD-level)

- Analyzed Pessimistic Value Iteration (PEVI) for offline RL, deriving clean error bounds for general and linear MDPs.
- Introduced principled simplifications using ε -covering numbers and self-normalized concentration.
- Produced a rigorous final report demonstrating provable efficiency of PEVI under realistic assumptions.

Multispeaker Transcription

Hsinchu, 2023

Hackathon Project, Meichu Hackathon (Google Group)

- Built an Android app with Python backend server capable of identifying active speakers, transcribing multi-speaker audio in real time.
- Integrated LLM-based language refinement to improve readability of segmented transcripts, delivering a smoother user experience.
- Recognized with third place(Bronze Award) in the Google Accessibility Group category among 10+ participating teams.

Honors & Awards

Bronze Medal, 2022 ICPC Asia Taoyuan Regional Programming Contest

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Bronze Medal, 2023 Meichu Hackathon (Google Group Android Accessibility)